

# AI for skin cancer: the ISIC community

A short briefing for attendees of the International Skin Imaging Collaboration (ISIC) meeting, September 2026.

|                                                        |                                     |                                                        |                                   |                                                          |
|--------------------------------------------------------|-------------------------------------|--------------------------------------------------------|-----------------------------------|----------------------------------------------------------|
| <b>553,019</b><br>public images in the<br>ISIC Archive | <b>1,207,517</b><br>images in total | <b>62</b><br>collections, about <b>22</b><br>with DOIs | <b>45</b><br>researchers profiled | <b>278</b><br>shared AI and imaging<br>papers, 2023–2026 |
|--------------------------------------------------------|-------------------------------------|--------------------------------------------------------|-----------------------------------|----------------------------------------------------------|

## ■ At a glance

For the past decade, the International Skin Imaging Collaboration (ISIC) has driven dermatology AI research by curating open benchmark datasets and running competitive image-classification challenges. Recent challenges have shifted toward releasing winning model weights and code directly to the community, giving machine learning teams reproducible state-of-the-art baselines and providing clinical investigators inspectable tools for validation studies.

Three recent milestones highlight this shift. First, the SLICE-3D dataset and the ISIC 2024 challenge established 3D total-body photography (3D-TBP) as an open benchmark, showing that incorporating patient-level spatial context improves lesion triage over single-lesion classification [1, 2]. Second, PanDerm demonstrated the utility of a shared dermatology foundation model evaluated across four distinct imaging modalities in clinician-in-the-loop studies [3]. Third, investigators published the first randomized controlled trial evaluating 3D-TBP in clinical practice alongside an empirical cost-effectiveness analysis [4, 5].

Recent community studies also outline where current systems fail [6–10]. Further progress will require shifting metrics from narrow diagnostic classification toward actionable management recommendations, integrating multimodal patient data, and ensuring calibrated performance across varied skin types and acquisition hardware. While the ISIC Archive now hosts the patient-level, longitudinal, and multimodal records needed to tackle these challenges, these capabilities remain underused across the research community.

This overview outlines recent research milestones across the International Skin Imaging Collaboration (ISIC), details active research priorities among consortium investigators, and directs readers to the appropriate leads for prospective partnerships. By highlighting underused archive capabilities, including longitudinal tracking, multimodal image pairing, and content-based image retrieval, we invite external groups to contribute datasets and join active studies supporting standardized, early skin cancer detection.

## ■ 1. Recent progress

**3D total-body photography, from dataset to trial.** The ISIC AI Working Group initiated SLICE-3D, 400,000 lesion crops from Canfield Vectra WB360 scans at seven institutions [1], which became the ISIC 2024 Kaggle challenge (2,739 teams) [11]. The winning model reached an AUC of 0.967 and, at 80% sensitivity, flagged about one cancer per 52 triaged lesions against one per 1,100 at baseline [2]. The study showed that patient context (the "ugly duckling" sign) was effective in the winning model. Queensland published the first randomized trial of 3D-TBP in high-risk patients with a cost-effectiveness analysis [4, 5], and the iToBoS consortium released 16,954 3D-TBP images from 100 participants [12]. Barcelona has tested an autonomous total-body and dermoscopy device [13], Girona is building a multi-robot scanner [14], and MSKCC has begun the first clinical assessment of the ISIC 2024 triage algorithm on real 3D-TBP data [15].

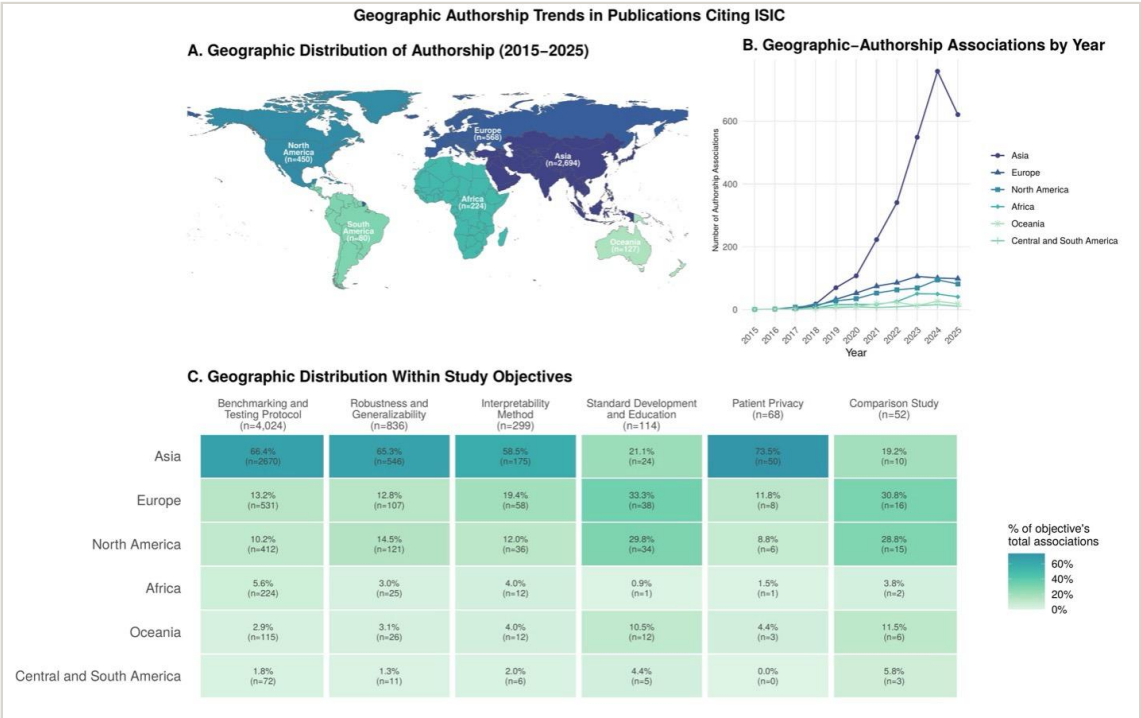
**Foundation models, vision-language models and image search.** PanDerm (Monash with Vienna, Queensland and Sydney) [3] was pretrained on 2.1 million images from 11 institutions across four imaging modalities. It outperformed clinicians in early melanoma detection from longitudinal images, improved dermatologists' dermoscopy accuracy by 11%, and improved non-specialists' differential diagnosis by 16.5% across 128 conditions. MSKCC and Northeastern fine-tuned a multimodal large language model to predict 16 quantitative lesion attributes from SLICE-3D images ( $R^2$  0.71–0.96) and showed that its representations support attribute-aware image retrieval [16]. This model now powers content-based image retrieval in the ISIC Archive (Section 4). Microsoft's general medical models [17, 18] raise the question whether dermatology-specific models still beat general ones.

**Human-AI collaboration and explainability.** A reinforcement-learning model that encodes clinician and patient preferences [19] raised melanoma sensitivity from 61.4% to 79.5% and improved dermatologists' correct diagnoses by 12%, by targeting the management decision rather than the label. Explainability is now tested on people: eye-tracking and reader studies at MSKCC [20] and in Heidelberg with Vienna [21, 22], and a 2026 study showing that explanations help clinicians and lay users in different ways [23].

**Data quality, fairness and new modalities.** New curated datasets (MILK10k, MIDAS, BALD, a longitudinal 3D-TBP set, ISIC-DICM-17K) [24–28] arrived with ISIC-DX terminology [29]. MSKCC compared skin-tone scales prospectively [30] and, with Northeastern, showed that top ISIC 2020 models were poorly calibrated across various facets, such as sex, skin tone and age [8]. Recently, pilot reflectance confocal microscopy (RCM) data have been uploaded to ISIC and can now be visualized with an integrated multimodal viewer. Beyond RCM, line-field confocal OCT (LC-OCT) is another imaging modality that has been increasingly studied in the field [31]. However, LC-OCT data have not yet been ingested into the ISIC Archive.

**Real-world performance.** Studies in routine care show how AI behaves outside curated test sets. In MSKCC's prospective PROVE-AI study, an open-source dermoscopy algorithm assessed 603 lesions, 95 of them melanomas [32]. Set to miss almost no melanomas, it reached 96.8% sensitivity and, with 37.4% specificity, saved one-third of benign biopsies. In Zürich, a convolutional neural network gave inconsistent results when the same lesion was imaged repeatedly in routine care [33], and a study from Athens concluded that image-based AI is not sufficient on its own to assess skin tumors in real clinical practice [34]. A 2026 systematic review and meta-analysis now pools the prospective evidence on AI-assisted melanoma diagnosis [35]. Early studies of language models as diagnostic readers point the same way: in a Heidelberg study with Vienna, they improved physicians' accuracy but also encouraged false reliance on their answers [36].

**Scoping review.** We conducted a scoping review guided by JBI methodology and PRISMA-ScR, searching five databases for original studies published from 2013 through August 2025 that used ISIC datasets for dermatologic AI research to understand the impact that the public ISIC Archive has had on the field. Of 5,194 records screened, 3,468 studies were included and verified as using ISIC datasets, which were used most often for training and internal validation, while external testing was substantially less common across the cohort. Model architectures evolved from classical machine learning to convolutional neural networks and, more recently, transformers and hybrid models. Authorship was concentrated in Asia, accounting for 65% of author-geography associations, whereas Europe and North America contributed the largest shares of ISIC datasets (Figure 1). This highlights how public datasets such as the ISIC Archive can have broad-ranging global impact. The review paper will soon be submitted for publication.



**Figure 1.** Geographic and temporal trends in authorship and study objectives. (A) Overall distribution of author affiliations by geographic region; publications with authors from multiple regions contributed to each author-geographic association, so counts reflect associations rather than unique publications. (B) Number of geographic-authorship publications each year from 2015–2025.\* (C) Distribution of geographic-authorship associations within each study objective; bars represent one study objective and display the proportion of geographic-author associations contributed by each geographic region within that objective. Counts within each study objective were converted to proportions so that each cell represents the proportion of all geographic-study objective associations attributed to a given continent. \*Data were collected only through August 2025 and are not representative of the full calendar year.

## 2. Where the field is heading

Nine themes dominate current work. **3D-TBP and whole-patient triage** is the most mature (Queensland, MSKCC, Barcelona) [1, 2, 4, 37]; device independence and longitudinal change detection remain open [2, 27, 38], and the only cost-effectiveness analysis so far is Australian [5]. For **foundation, vision-language and agentic models** (Monash, Vienna, Queensland, Stanford, industry) [3, 18, 39–42] the question is now whether they help clinicians and generalize [7]. **Human-AI interaction and explainability tested on people/readers** is led by Vienna, Lisbon, MSKCC, Heidelberg and Zürich [19–23, 43]; **data quality and transparency** by Basel/Lucerne, Manchester, MSKCC and Vienna [24, 29, 44–47]; **fairness and skin tone** by MSKCC, Stanford and Queensland, with measurement moving from Fitzpatrick self-report toward Monk scales, colorimetry and scanner-derived color [8, 30, 48–50]. **New optical modalities** (RCM, LC-OCT, OCT angiography, multispectral and fluorescence imaging) are pursued at MSKCC, Stanford, Barcelona, Sydney, Vienna, Zürich, Arizona and Oregon [31, 51–57]. **Large language models** as readers, scribes and patient-information sources are evaluated at Stanford, Heidelberg, Santiago, MD Anderson and Queensland [36, 58–62]. A smaller cluster works on **robustness, security and data standards** (Kraków, Manchester, Lisbon with Campinas, Vitória, Vanderbilt, Seoul) [28, 63–68]. Finally, **what AI does not fix** is a theme of its own: two 2026 studies from Arizona, Oregon and Stanford showed that a val-

idated early-detection toolkit improved accuracy without reducing the number needed to biopsy, and that much biopsy-decision variability cannot be predicted by AI [9, 10]. With the "reality check" and "limits" papers [6, 7], this points to the management decision as the next target.

### 3. The ISIC Archive

The Archive's 62 collections (1,207,517 images in total, 553,019 public; more than 12,000 registered users) hold [69] contact and polarized dermoscopy, clinical close-up and overview photographs, 3D-TBP tiles, segmentation masks and longitudinal series, with diagnoses on the ISIC-DX tree and a Creative Commons license on every image. It can be browsed in the gallery, searched by image, or downloaded through the API or `isic-cli`. The ISIC Archive is listed on the AWS Open Data registry, which adds long-term sustainability. It is the field's common benchmark, is included in the pretraining corpus of foundation models such as Pan-Derm [3], and a test bed for fairness and data-quality audits (more than 2,500 citations and 3,500 challenge participants). Users have provided extensive feedback on the Archive through their studies, helping with quality assurance by finding duplicates and leakage in the older challenge sets [46], shortcut artifacts such as rulers and ink, and under-representation of darker skin tones.

Table 1. Major public collections in the ISIC Archive.

| Collection                                       | Images  | Notes                                                                   |
|--------------------------------------------------|---------|-------------------------------------------------------------------------|
| ISIC 2024 challenge / SLICE-3D                   | 401,059 | 3D-TBP crops from seven institutions; permissive subset available       |
| BCN20000                                         | 18,946  | Hospital Clínic Barcelona dermoscopy                                    |
| Melanoma and nevus with confirmed histopathology | 18,133  | DOI 10.34970/277003                                                     |
| ISIC-DICM-17K                                    | 17,060  | Balanced melanoma/non-melanoma set (Manchester)                         |
| iToBoS 2024                                      | 16,954  | 3D-TBP skin regions, 100 participants, bounding boxes                   |
| IMA++ multi-annotator segmentation               | 14,967  | 17,684 masks from 16 annotators                                         |
| DERM12345                                        | 12,345  | Türkiye; 40 subclasses                                                  |
| HAM10000                                         | 11,720  | Vienna and Queensland                                                   |
| MILK10k                                          | 10,480  | Paired dermoscopy and clinical; 48 diagnoses; skin tone; MONET concepts |
| MSKCC skin-tone labeling                         | 4,879   | Fitzpatrick, Monk, Pantone and colorimeter                              |
| MEL-SELF                                         | 3,008   | Patient self-surveillance trial                                         |
| BALD                                             | 666     | Acral lesions                                                           |

**Getting a DOI.** Standardized benchmarks are essential for evaluating dermatology models, and the ISIC Archive serves as the primary repository for this data. Contributors prepare images and metadata against the ISIC data dictionary, upload them through the web interface or `isic-cli` as a collection, and assign a Creative Commons license to each image; ISIC staff review for privacy and quality, then mint a DataCite DOI (prefix 10.34970) against a frozen snapshot. About 22 collections carry DOIs, including SLICE-3D (the ISIC 2024 challenge data), MILK10k, DERM12345, iToBoS 2024 and others.

### 4. Underused resources in the Archive

The vast majority of studies built on ISIC data use only the raw lesion crop and its primary diagnostic label. In contrast, the current ISIC Data Dictionary (version 2.4, April 2026) [70] specifies granular patient-level, anatomical, and acquisition metadata. Paired with the archive's image-similarity search capabilities, this expanded schema provides the foundation for longitudinal lesion tracking, cross-modal analysis, and patient-contextual modeling, avenues of investigation that few research teams have pursued.

Table 2. Underused field groups in data dictionary v2.4 and what they make possible.

| Field group                      | Fields                                                                                                                                                              | What they make possible                                                                                                                                |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Provenance and licensing         | <code>image_manipulation</code> , <code>copyright_license</code>                                                                                                    | Excluding synthetic or edited images; building training sets that respect commercial and non-commercial licenses                                       |
| Patient context                  | <code>patient_id</code> , <code>age_approx</code> , <code>sex</code> , <code>fitzpatrick_skin_type</code> , <code>personal_hx_mm</code> , <code>family_hx_mm</code> | Comparing a lesion with the same patient's other lesions; adjusting for personal and family history; auditing performance by age, sex and skin type    |
| Lesion timeline                  | <code>lesion_id</code> , <code>acquisition_day</code>                                                                                                               | Modeling change over time from repeated images of the same lesion                                                                                      |
| Modality and optics              | <code>image_type</code> , <code>dermoscopic_type</code> , <code>tbp_tile_type</code> , <code>rcm_case_id</code>                                                     | Pairing clinical, dermoscopic, 3D-TBP and confocal views; combining white-light and cross-polarized tiles; aligning the confocal images of one session |
| Histopathology and label quality | <code>concomitant_biopsy</code> , <code>diagnosis_confirm_type</code> , <code>mel_thick_mm</code> , <code>mel_mitotic_index</code> , <code>mel_ulcer</code>         | Weighting labels by how they were confirmed; predicting prognostic features rather than only a diagnosis                                               |

**Patient context and lesion timelines.** `patient_id` links a person's images within an upload cohort, so a model can score each lesion against that patient's own background, the computational form of the "ugly duckling" sign that improved triage in ISIC 2024 [2]. `lesion_id` with `acquisition_day` turns static images into sequences for models that learn from change, as sequential digital dermoscopy does clinically. Patient and lesion identifiers are unique, so records can be linked across collections.

**Modality, optics and label quality.** `image_type` covers dermoscopy, clinical views, 3D-TBP tiles and three confocal types; `tbp_tile_type` separates white-light from cross-polarized tiles, and `rcm_case_id` groups the confocal images of one session. The schema is ready for RCM even though no public RCM collection has yet been contributed. `concomitant_biopsy` marks images taken at or just before biopsy; `diagnosis_confirm_type` records which of five methods confirmed a label; and Breslow thickness, mitotic index and ulceration allow models to predict prognostic features, not only a diagnosis. `image_manipulation` flags altered and synthetic images, and per-image licenses allow commercial and academic training sets to be filtered programmatically.

**Search by image.** The Archive's content-based image retrieval (CBIR) is built on the attribute-grounded model of Torop et al. [16]. Starting from an image, users retrieve visually similar images; the model can also specialize the search to a quantitative attribute such as lesion area, so results match both appearance and measurement. This makes it easy to assemble look-alike sets for read-studies and teaching, or to check new data against what the Archive already holds.

**Limits to know.** The ISIC Archive currently supports only the structured data fields described in the data dictionary. Contributors who want to share data beyond these fields can still do so through DOI sidecar files; however, information in those extra fields can be accessed only through the collection, not through the gallery's faceted search or the API.

## 5. Funding

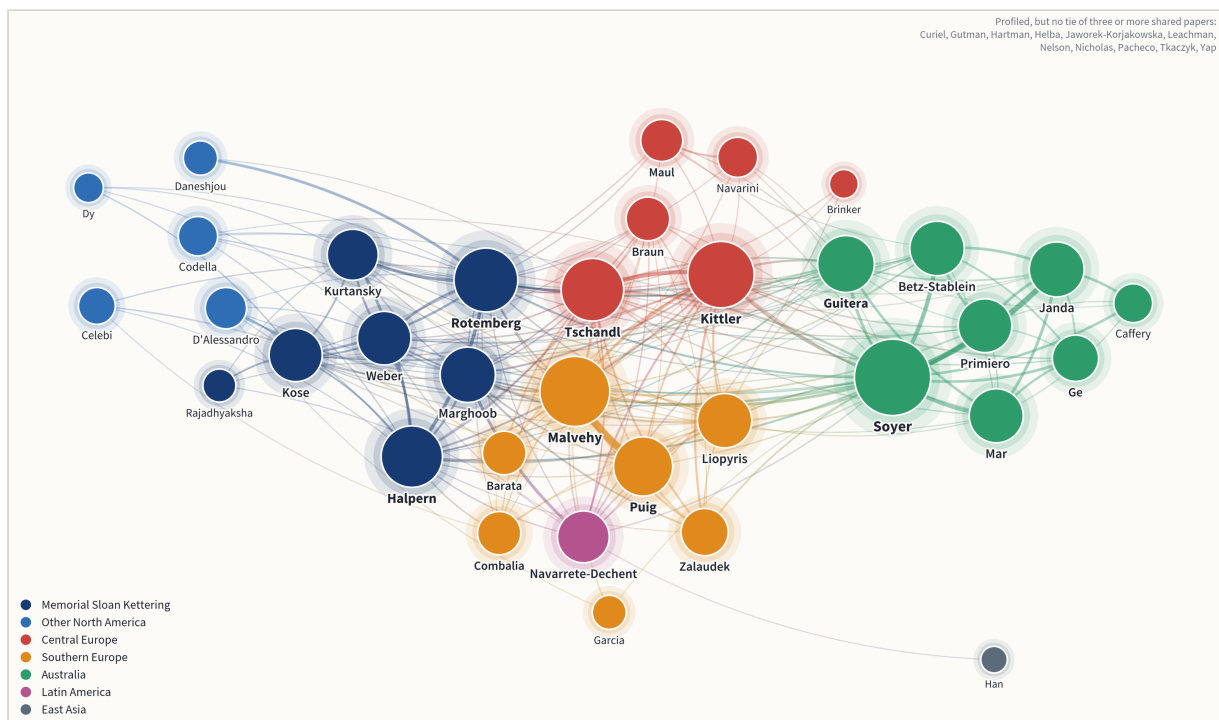
The Archive's infrastructure is funded by two NIH/NCI U24 grants to MSKCC and Shore Family philanthropic funding. So far, the science around ISIC research draws on the funders listed below (to our knowledge).

Table 3. Funding related to ISIC and its members.

| Funder                                | Grant or program                 | Purpose and lead                                                                               |
|---------------------------------------|----------------------------------|------------------------------------------------------------------------------------------------|
| NIH/NCI                               | U24CA264369                      | Multimodal-ISIC platform; Rotemberg                                                            |
| NIH/NCI (ITCR)                        | U24CA285296                      | ISIC-REPO Archive FAIR compatibility; Halpern                                                  |
| NIH/NCI                               | R01CA240771, R01CA199673         | RCM and AI; Rajadhyaksha                                                                       |
| US DoD (CDMRP)                        | HT94252410552                    | Collection of TBP crops and TBP-plus-dermoscopy model development; Halpern                     |
| US DoD (CDMRP)                        | Melanoma Academy Scholar         | FIND-MEL, following nevi to detect melanoma; Rotemberg                                         |
| Melanoma Research Alliance            | Young Investigator Award         | Rotemberg                                                                                      |
| Melanoma Research Alliance            | Team Science Project             | Collection of smartphone images and AI model development; Halpern                              |
| EU Horizon 2020                       | 965221, iToBoS                   | 3D total-body scanner and AI; Univ. Girona with Barcelona, Queensland, Trieste, IBM and others |
| EU Horizon Europe                     | 101096667, MELCAYA               | Melanoma in children, adolescents and young adults; Puig                                       |
| Australian Cancer Research Foundation | ACEMID                           | In 2018 for 15 Vectra WB360 systems; Soyer                                                     |
| NHMRC                                 | 2001517                          | ACEMID cohort; Mar                                                                             |
| NHMRC                                 | 2006551, 2009923, APP1135285     | Centre of Research Excellence and Synergy grants; Janda                                        |
| UK EPSRC                              | EP/N02700/1                      | ISIC data-quality analysis; Yap                                                                |
| Portugal FCT                          | CEECIND/00326/2017               | Barata                                                                                         |
| In-kind                               | Kaggle, AWS, Canfield Scientific | Challenge hosting, data hosting, workshop sponsorship                                          |

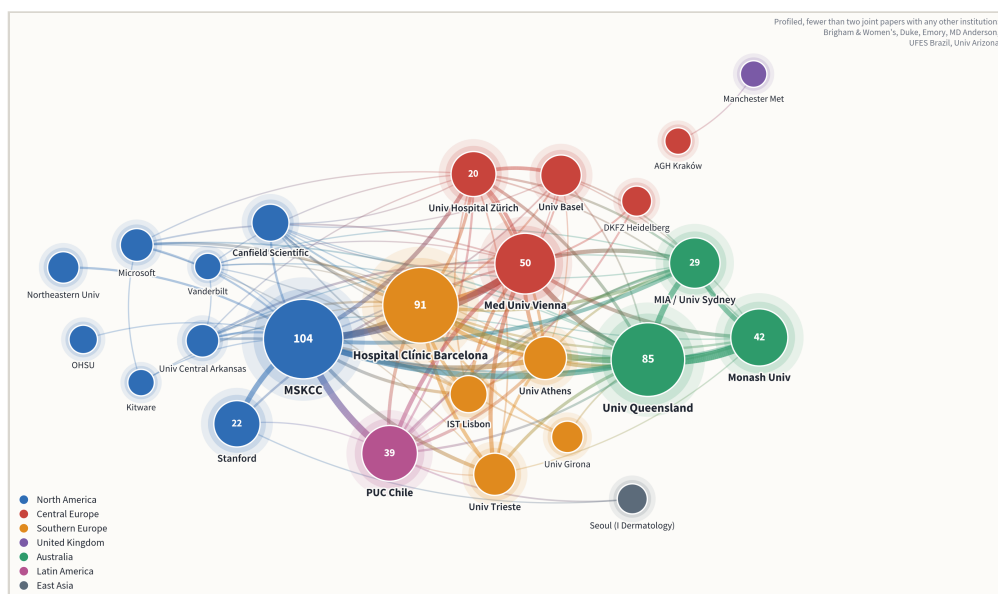
## 6. Who works with whom

We cross-indexed the full 2023–2026 publication lists of the 45 profiled researchers (Organizing Committee, AI Working Group, other working-group leads and six other groups connected to ISIC), giving 278 papers on AI, imaging or digital health with two or more network members (method in Appendix A).



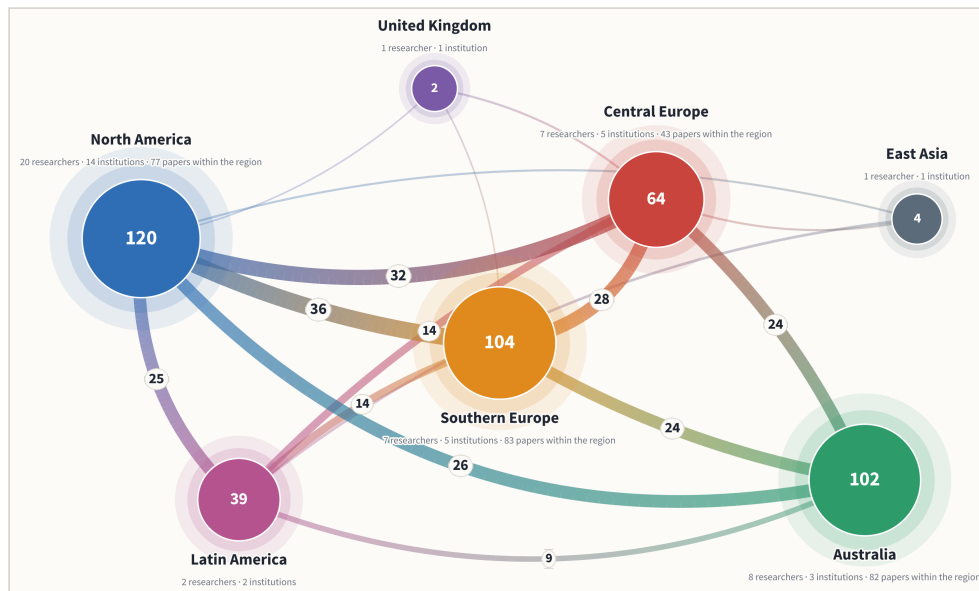
**Figure 2.** Co-authorship among researchers, 2023–2026. Node size is total shared papers; line width is papers shared by the pair (ties of three or more shown).

The network has four hubs of similar weight. Queensland is the densest, held together by the ACEMID program (Soyer shares 44 papers with Janda, 30 with Betz-Stablein). In Barcelona, Malvey and Puig share 67 papers and act as one unit. Vienna (Kittler and Tschandi, 30) and MSKCC (Halpern and Rotemberg, 23) complete the core. Links between hubs run mainly through Kittler, Soyer and Malvey. MSKCC has fewer joint papers with each of the other hubs than they have with each other. Its central place comes from the Archive, the challenges and consensus papers, not from joint clinical studies. Within MSKCC, Kose is tied to informatics, optics and clinical dermoscopy, and one of ISIC's strong links to an external machine-learning lab (Northeastern). Several profiled researchers with few ties run active programs on robustness, validation, data standards and image quality that the core does not yet use.



**Figure 3.** Joint papers between institutions. The number in each circle is papers involving that institution.

By institution (Figure 3) the heaviest links are Monash–Queensland (33), MSKCC–Vienna (24), MSKCC–Barcelona (21) and Barcelona–Vienna (21). Another big contributor is PUC Chile, whose prominence rests on one researcher.



**Figure 4.** Papers co-authored across regions. Positions are schematic.

By region (Figure 4) the strongest links are North America–Southern Europe (36), North America–Central Europe (32), Central Europe–Southern Europe (28) and North America–Australia (26). The United Kingdom and East Asia are almost absent despite having AI Working Group members; Latin America's apparent strength rests on one researcher; and Africa, South Asia and the Middle East do not appear, although partners exist there (Basel's PASSION dataset from sub-Saharan Africa [71] and MSKCC's RCM work in India [72]).

## 7. Upcoming grant proposals and collaborations

### Evidence-grounded vision-language AI for interpretable skin cancer diagnosis

**Call:** NIH R01 // **Partners:** MSKCC, Northeastern, Mayo Clinic

Dermatology lacks a structured, public infrastructure for morphology-grounded AI. Current archives and ontologies like D3X lack localized dermoscopic annotations linked to clinical vernacular and diagnostic reasoning. We propose building an evidence-weighted annotation layer within the International Skin Imaging Collaboration (ISIC) platform to train, audit, and evaluate an explainable vision-language model (DermVLM<sup>2</sup>). We hypothesize that anchoring diagnostic predictions to inspectable, localized morphology makes AI safer and more clinically actionable.

- **Aim 1:** Construct a Morphologic Knowledge Schema (MKS) and release ~10,000 expert-adjudicated, spatially localized annotations via ISIC.
- **Aim 2:** Train DermVLM<sup>2</sup> to jointly predict diagnoses and localized morphology, audit reasoning via a Logic-Alignment Score, and generate ~250,000 auto-labeled images.
- **Aim 3:** Prospectively test whether morphology-grounded explanations improve clinician confidence calibration, management decisions, and resistance to automation bias over standard AI.

### Multimodal spatial-temporal AI for lesion-level melanoma triage

**Call:** Model-to-Clinic (M2C) for Precision Medicine with AI: Integrating Imaging with Multimodal Data (PRIMED-AI) (UG3/UH3, Clinical Trial Optional) // **Partners:** MSKCC, Stanford University.

Longitudinal melanoma surveillance in high-risk patients requires manually screening hundreds of nevi using 3D total-body photography (TBP) and serial dermoscopy. Current deep learning approaches evaluate cropped lesions in isolation, ignoring patient-level phenotype, spatial context (the "ugly duckling" sign), and temporal change. Consequently, these models suffer from poor calibration and high false-positive rates that limit clinical utility. We propose the Context-Aware Lesion Triage Clinical Decision Support (CDS) framework, a multimodal spatial-temporal AI system designed to align with clinical reasoning. In the UG3 phase, we will:

- **Aim 1:** Build a unified multimodal pipeline harmonizing TBP crops, dermoscopy, and clinical covariates for standardized data ingestion and inference.
- **Aim 2:** Develop a spatially grounded triage model using patient-specific lesion graphs and graph attention networks to operationalize the "ugly duckling" principle.
- **Aim 3:** Construct longitudinal alignment pipelines and temporal embeddings to track individual lesion evolution and phenotypic drift across serial visits.

In the subsequent UH3 phase, we will integrate the validated system into clinical enterprise imaging software (e.g., Canfield Vectra via API), execute prospective clinical trials, and partner with the PRIMED-AI Validation Center for independent evaluation, delivering an interpretable, scalable surveillance tool.

### **Engagement Platform**

Developing robust dermatologic vision-language models requires fine-grained, spatially localized annotations that are expensive and slow to produce by hand. The Engagement Platform extracts these labels directly from expert case reviews by synchronizing interactive annotation tools, real-time mouse tracking, and speech transcription against baseline foundation model proposals.

Prior to each review, the platform populates candidate diagnoses and lesion-level morphologic boundaries using pre-trained dermatology models. As an expert inspects the image, the software aligns the clinician's spoken dialogue with high-resolution cursor movements. This spatial-audio alignment resolves spatial references (such as pointing while naming an atypical pigment network or blue-white veil) and automatically confirms or refutes the model's proposals. The clinician can add direct manual annotations, while an automated check flags any unaddressed candidate findings for brief sign-off before finalizing the case.

The resulting adjudicated cases and spatial interaction logs serve a dual purpose within an adaptive learning environment for trainees. The platform monitors resident cursor navigation, diagnostic choices, and misclassification patterns, using these metrics to detect specific diagnostic blind spots and dynamically serve follow-up cases targeting those errors. Integrated with the International Skin Imaging Collaboration archive, the platform turns routine expert case reviews into a stream of spatially grounded annotations while systematically personalizing clinical instruction for novices.

## **8. How to get involved**

Contributors can make their collections far more useful by filling in the patient, lesion-timeline, modality and histopathology fields, and by contributing modalities the Archive lacks, such as RCM and LC-OCT. Researchers can use what is already there: patient context, lesion sequences, confirmation methods and image search are open for study. Collaborators are most needed where the network is thin: groups working on robustness, data quality, image acquisition and data standards, and regions that are absent or represented by a single researcher.

## **Appendix A. How the collaboration network was built**

Publication lists were collected for the 45 profiled researchers (40 Google Scholar profiles, four Scopus author records and one ORCID record); one further researcher (Maul) was added from a PubMed search, so the individual network in Figure 2 can show up to 46 people. Titles from 2023–2026 were normalized and cross-indexed, and two people were linked whenever a paper appeared on both of their lists or a verified author list named both. Prefaces, corrections and replies were excluded, and only papers whose titles concern AI, imaging or digital health were kept (278 papers). Institutions and regions were aggregated so that each paper counts once per institution, region or pair.

The network records co-authorship, not shared data, grants or informal collaboration. Conference abstracts are counted, so pairs that publish an abstract and then a paper are slightly inflated; Google Scholar truncates long author lists, so authorship position is not used; and the topic filter is keyword-based and approximate. Affiliations follow each researcher's profile.

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